

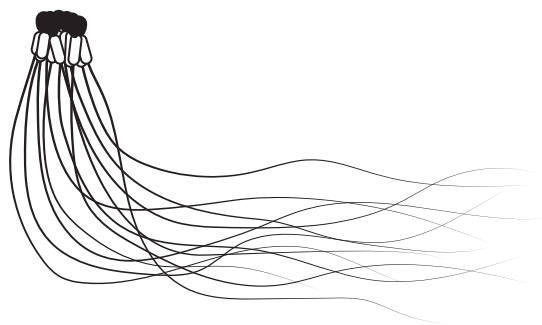
8 Gametes are specialised for their function.

Egyptian female desert ants (*Cataglyphis savignyi*) mate with more than one male.

A male desert ant has an adaptation to improve the chances of his sperm fertilising a female's egg cell.

The sperm are released in bundles with the heads held together by a sticky protein.

The diagram shows a bundle of desert ant sperm.



(a) Suggest how the sticky proteins would have been secreted.

(2)

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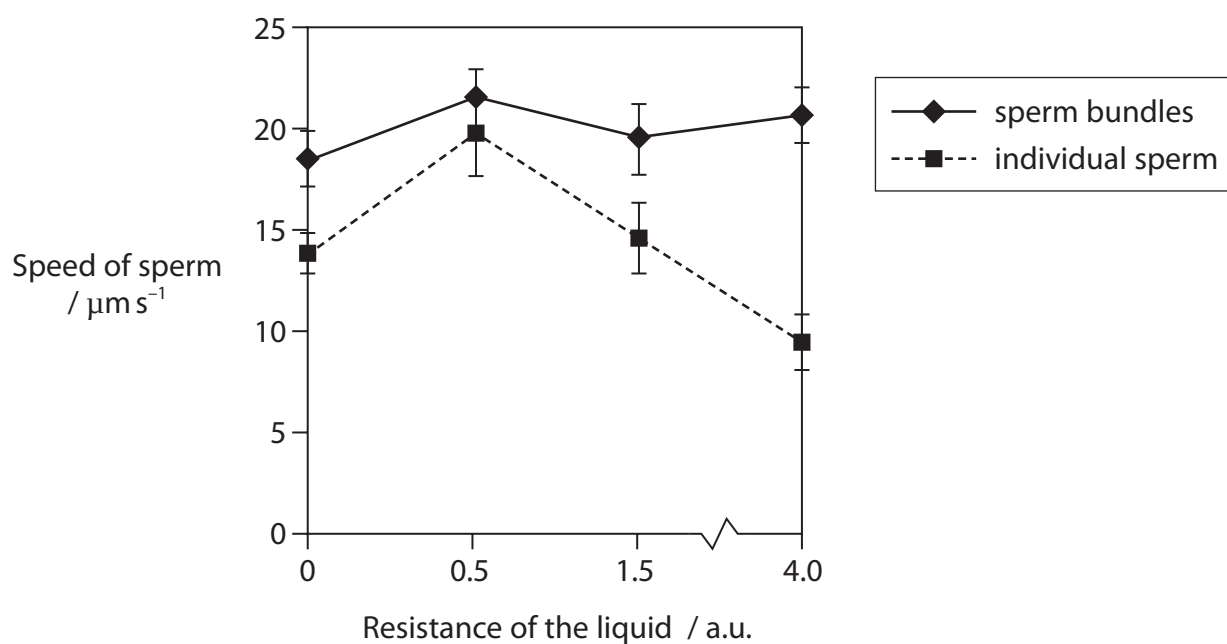
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(b) The graph shows the speed of individual sperm compared with bundles of sperm in liquids of different resistance.



Comment on the results of this investigation.

(4)



- (c) The photograph shows an Amur leopard found in southeastern Russia and northern China.

In-vitro fertilisation (IVF) techniques are being developed to help conserve this endangered species of leopard.



(Source: © NaturalLight / Alamy Stock Photo)

These techniques are being developed using mice.

IVF involves the removal of egg cells from the female mice. The zona pellucida can be damaged during this process.

The effect of damaging the zona pellucida was investigated.

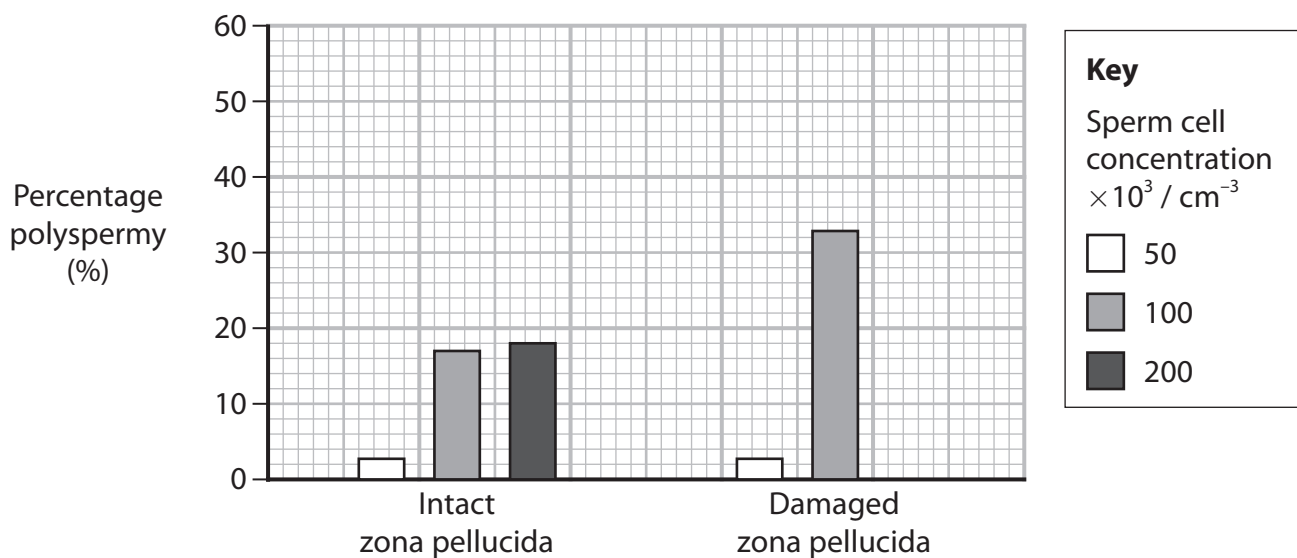
Three hundred mouse egg cells were distributed equally in three Petri dishes containing nutrient solutions.

Samples of sperm of different concentrations were added to each Petri dish. The volume of each sample added was the same.

The experiment was repeated for egg cells which had their zona pellucida damaged.

The graph shows the percentage of egg cells fertilised by more than one sperm (percentage polyspermy).





- (i) When the highest sperm concentration was added to the egg cells with damaged zona pellucida, the percentage polyspermy was 52%.

Plot this result on the graph.

(1)

- (ii) Explain the results of this investigation.

(4)

(Total for Question 8 = 11 marks)